V = velocity, M = mass of proton, B = magnetic field
q = charge of particle, $K.E$ = kinetic energy of charged particle.

Ref. No. Coils-1
Date

- Assumptions
  - Considering proton as the charged particle for my design

  - All this calculations are for type-1
  - considering R_{in} (inner radius of vaccum chamber)
    R_{ex} (external radius of vaccum chamber)
    $R_{in} = 0.15$ M
    $R_{ex} = 0.5$ M

  - considering minimum K.E required of proton to cause fission of U-238 : 40 MeV
    (since proton is a charged particle the minimum K.E required is significantly higher than neutrons)

- Calculations for B_{max} (max magnetic field required)

$R = \frac{MV}{qB}$, $K.E = \frac{1}{2}MV^2$

$$40 \text{ MeV} = \frac{1}{2}MV^2$$

$$40 \times 10^6 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 1.67 \times 10^{-27} \times V^2$$

$$7.66 \times 10^{15} = V^2$$

$$\boxed{V = 87.54 \times 10^6 \text{ m/s}}$$

$$R_{in} = \frac{MV}{qB}$$

$$0.15 = \frac{1.67 \times 10^{-27} \times 87.54 \times 10^6}{1.6 \times 10^{-19} \times B_{max}}$$

$$\boxed{B_{max} = 5.836 \text{ Tesla}} \rightarrow \text{achieveable}$$

Consider proton as the charged particle

Minimum K.E required = 2 MeV

$r = \frac{mv}{qB}$, $K.E = \frac{1}{2}mv^2$

$\Rightarrow 2\times10^6 \, eV = \frac{1}{2} m v^2$

$2\times10^6 \times 1.6\times10^{-19} \, J = \frac{1}{2} \times 1.67 \times 10^{-27} \times v^2$

$3.83 \times 10^{14} = v^2$

$\boxed{v = 19.57 \times 10^6 \, m/s}$ ⟹ velocity of proton required to cause fission of U-238

$r = \frac{mv}{qB}$

$r = \frac{1.67\times10^{-27} \times 19.57\times10^6}{1.6\times10^{-19} \times B}$

$1 = \frac{0.2}{B}$

[assuming radius of setup to be 1 meter]

$\therefore B_{Max} = 0.2 \, T$

[The maximum magnetic field required]

[easily achievable]

The size can be made even smaller isn't it?

variable fix for proton

$$R = \frac{mv}{qB}$$

$mv \rightarrow$ variable (by changing nuclear core, particle)

$qB \rightarrow$ variable

fix for proton

- ~~If the B_{max} seems too high, we can reduce it by increasing the size of reactor ($\because R \propto \frac{1}{B}$) or By choosing nuclear fuel which doesn't require 40 MeV for fission.~~

- If B_{max} seems too high for practical application we can reduce it by,

1) Increasing the size of reactor.
   Since $R \propto \frac{1}{B}$

2) By choosing different nuclear fuel which doesn't require 40 MeV for fission.

3) By replacing proton with neutron, since nuclear repulsion increases the minimum energy required for fission significantly, but we would have to find a way to accelerate neutrons.

- The size can be made even smaller, isn't it? By ~~varying~~ changing the 3 factors mentioned above.